

Cloudflare Radar 2024 Year in Review

A data-driven analysis of Internet traffic, adoption trends, disruptions, and security threats in 2024



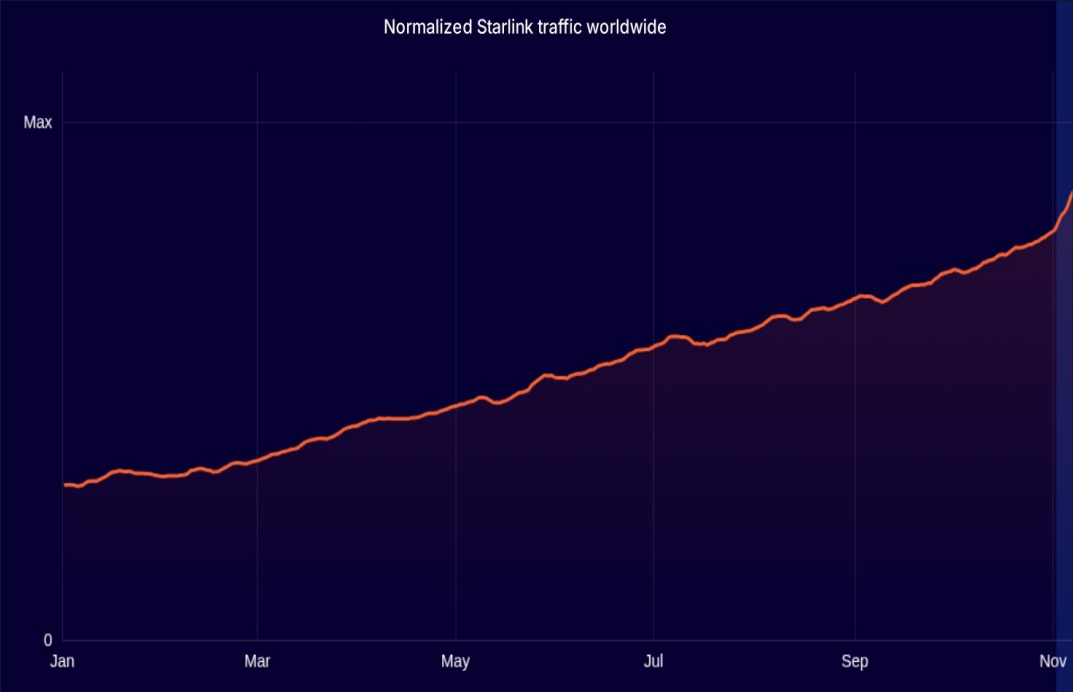
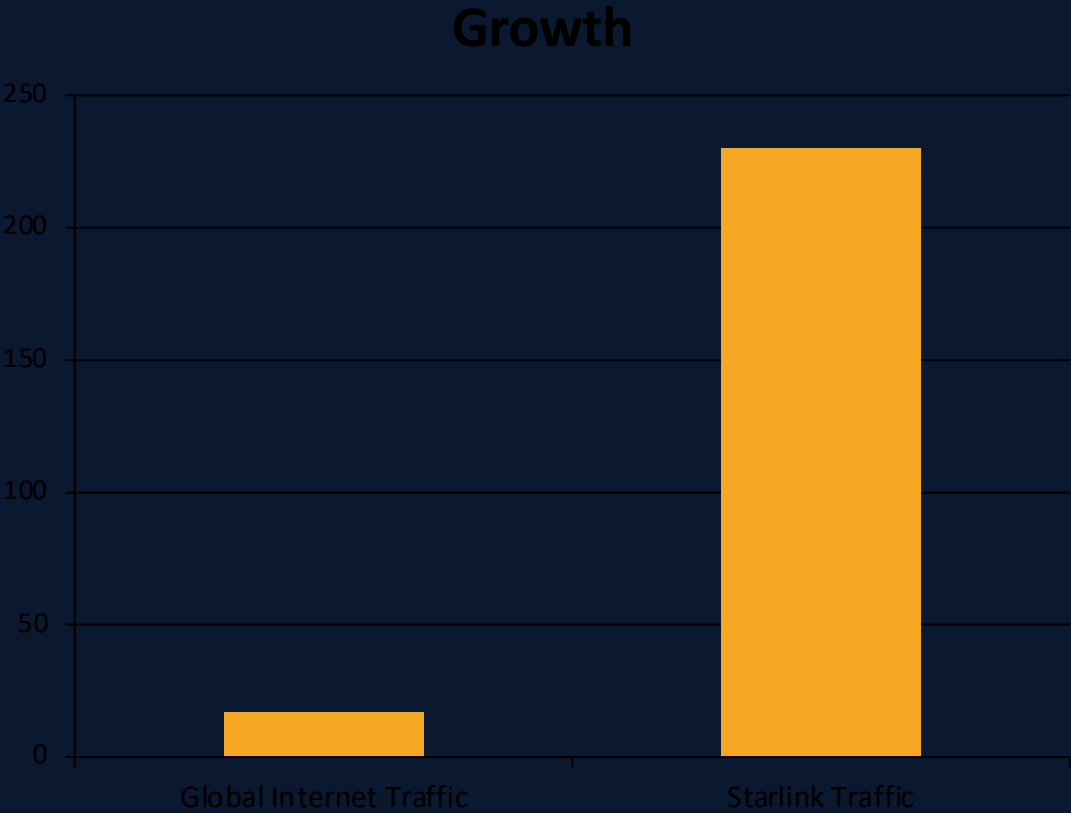
Traffic Growth

Technology Adoption

Disruptions

Security

17% Global Traffic Growth and Starlink's 3.3x Surge



- Global Internet traffic increased 17% year-over-year in 2024.
- SpaceX Starlink traffic surged 3.3x, expanding connectivity to underserved regions.
- Trend lines normalized from mid-January to remove post-holiday distortion.

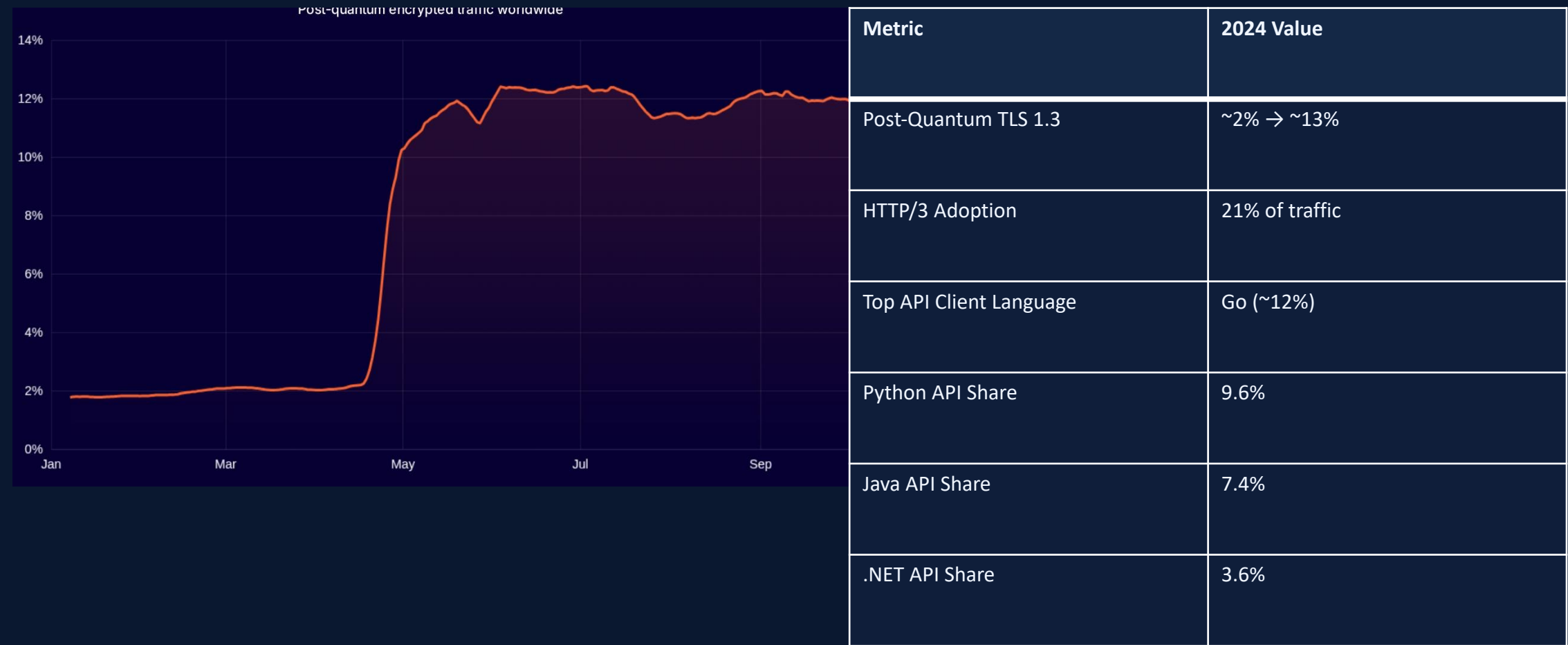
Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

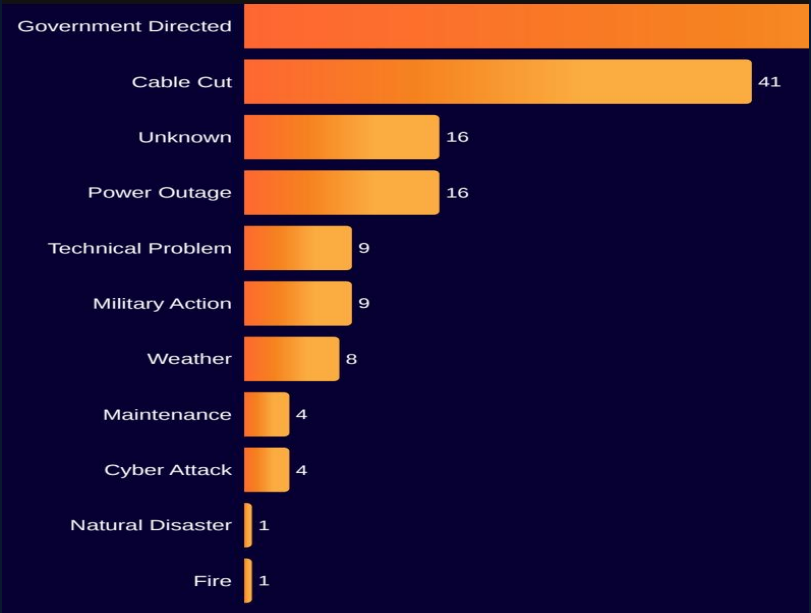
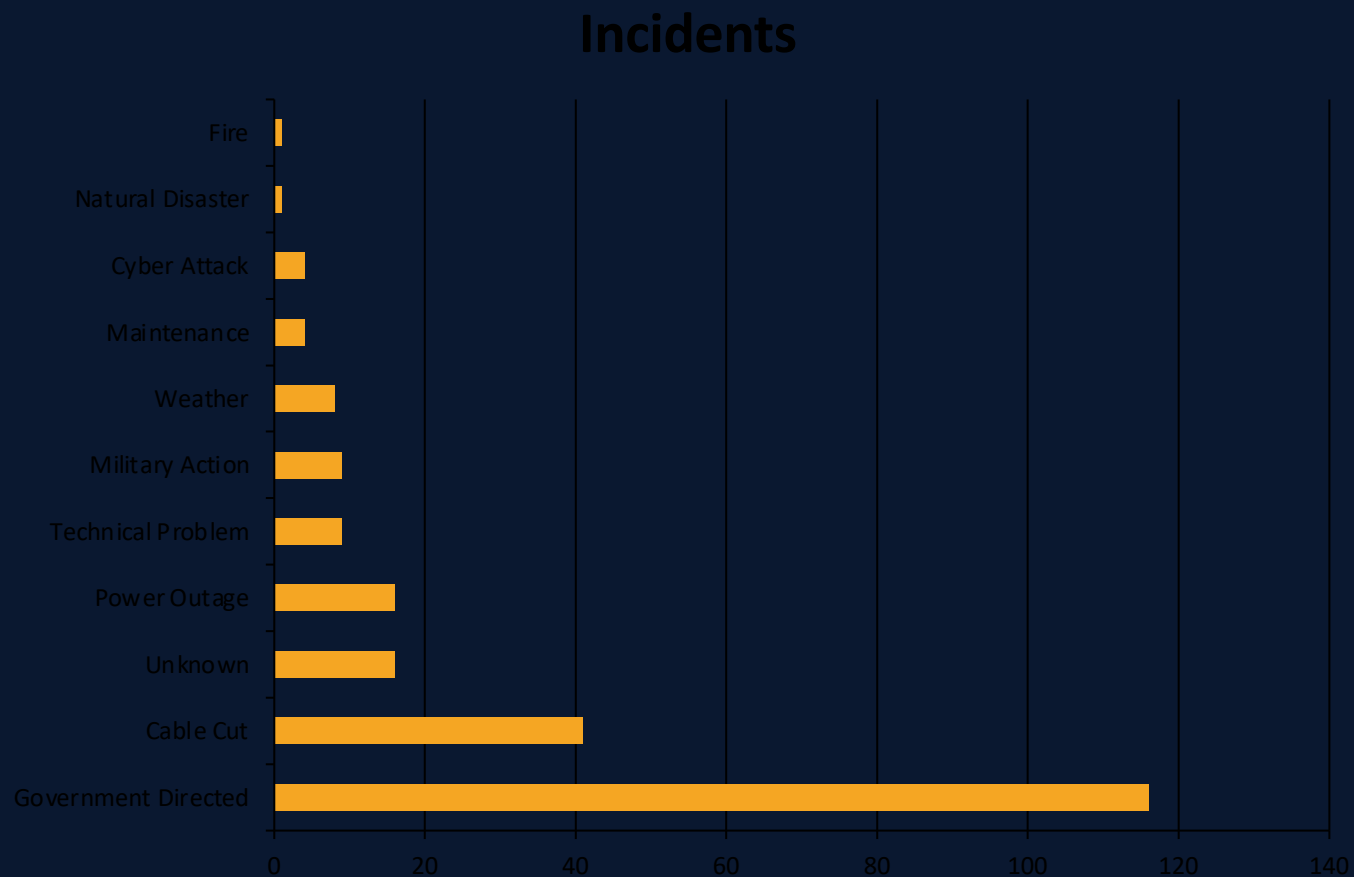
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Generative AI continued its rapid rise through 2024, while Metaverse momentum

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



225 Major Disruptions: Government Shutdowns Lead Outage Causes



- Total major disruptions observed: 225
- Government-directed shutdowns were the leading cause.
- Next highest: Cable cuts (41), Power outages (16).

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

- 6.5% of all traffic processed by Cloudflare was mitigated.
- Most-attacked sector: Gambling and Games.
- 68% of global bot traffic originated from top 10 countries.
- Log4j (CVE-2021-44228) remained actively targeted.
- RPKI valid routes increased by 6.4%.
- 4.3% of processed emails were malicious.
- 43% of malicious emails included deceptive links.
- .bar generated the largest share of malicious/spam email.

Five Takeaways Shaping the 2025 Internet Outlook

1. Demand trajectory remains strong: global Internet traffic grew 17% YoY, with new access models like Starlink scaling quickly (3.3x).
2. Post-quantum transition has entered mainstream deployment, climbing from ~2% to ~13% of TLS 1.3 traffic in one year.
3. Network resilience is increasingly policy-dependent: government-directed shutdowns were the top driver among 225 major disruptions.
4. Threat pressure remains persistent and broad-based: 6.5% mitigation rate, concentrated bot origins, and continued Log4j exploitation.
5. Protocol and developer ecosystem shifts (HTTP/3 at 21%, Go leading API clients) should inform 2025 platform and architecture decisions.